

Akshat Charan

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EDUCATION

•Bachelor of Technology in Computer Science and Engineering

Manipal University Jaipur

2020-24

CGPA: 8.52

EXPERIENCE

•Software Development Engineer

Statcon Powtech Private Limited

Jul 2024 -

- Revamped a WPF desktop application in C# that improved user task efficiency by 30%.
- Contributed to railway display board implementation with C and RTOS, improving real-time data accuracy by 40%.
- Developed a platform controller on Yocto Linux using QT and processed the data using TCP/IP protocols, reducing the overall project cost by 90%.
- Managed a TV application utilizing embedded Linux and the Qt framework, ensuring seamless integration and user-friendly operation.

•Software Intern

Statcon Electronics India Limited

Oct 2023 - Jun 2024

- Integrated a solar inverter with Wi-Fi connectivity, enabling real-time data transmission with 98% accuracy to a mobile app built on Flutter.
- Implemented protocols for ESP32 to ensure over 95% reliability in data communication.
- Streamlined STM32 firmware integration, reducing implementation time by 40%, allowing full control via the mobile app.
- Designed a user-friendly mobile app interface that improved usability and increased user satisfaction scores by 30%.

PERSONAL PROJECTS

•SENTIMENT ANALYSIS IN HEALTHCARE

- Published a research paper in ACM Digital Library, with over 100 downloads
- Utilized the UCI Machine Learning dataset to perform sentiment analysis on healthcare data, consisting of over 200,000 reviews.
- Obtained 92% accuracy with LSTM and 89% accuracy with LightGBM on the UCI ML dataset, outperforming traditional methods by 15%.

•SENTIMENT ANALYSIS ON TWITTER TWEETS

- Created a model to classify sentiments (positive, negative, neutral) with 90% accuracy, analyzing 50,000+ tweets.
- Evaluated API performance against traditional machine learning algorithm for sentiment analysis.

•VOICE ASSISTANT

- Developed a Python-based voice assistant that executed commands with 95% accuracy.
- Enhanced functionality with Selenium API, automating tasks and reducing manual operation time by 40%.

•WEATHER PREDICTION

- Applied ARIMA model for accurate weather prediction and forecasting for more than 5 Major Indian cities.
- Achieved RMSE below 3 across all targeted Indian cities and predicted data for the next 10 years.

TECHNICAL SKILLS AND INTERESTS

Languages and Frameworks: C/C++/C#, Python, QT

Additional Experience: TCP/IP, Dart, HTML+CSS, WPF, Flutter, RTOS

Dev Tools: Code Composer Studio, Arduino IDE, STM32CubeIDE, VScode, Git, Github

Hardware: ESP32, STM32, SBC

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Software Engineering.

Areas of Interest: Software Development, Embedded Software Development, Machine Learning.

Soft Skills: Self-learning, Presentation, Quick-Learner

Hobbies: Gym, Sports, Movies